Evaluating Expressions with Radicals

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Evaluate expressions containing square roots, cube roots and integer exponents, simplifying radicals by factoring within the benchmark's lists.

Must not Perfect squares up to 225 and perfect cubes from −125 to 125, or fractions whose parts are each in those lists. Estimating a root that is not perfect was Unit 2. No fractional exponents.

Target I can evaluate an expression that contains square roots, cube roots and integer exponents.

MTR **MTR.5.1** — Notes I · **MTR.6.1** — Boards 3 and 6 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The radical bar is a pair of parentheses you cannot see. Finish what is under it, then take the root.
2. A cube root of a negative number is ordinary. A square root of a negative number is not a number you have.
3. A negative exponent is still a reciprocal. Nothing about the radical changes that.

Slide by Slide

1. **Evaluating Expressions with Radicals.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 and Q4 are the two halves of today. *1 retrieves 4.04 — leading zeros never count; the trailing zero does, because a decimal point is written (yesterday). 2 retrieves 4.02–03 — multiply the coefficients, add the exponents (last week). 3 retrieves Unit 2 — perfect squares (2.01): 14 · 14 = 196 (last unit). 4 retrieves MA.5.AR.2.2 — multiply before you subtract; today the radical joins that list (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 3 2. 6×10^{-3} 3. 14 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy all three lines.** Whatever sits under the bar is inside a pair of parentheses you cannot see. Finish it, then take the root.
6. **Notes.** Cube roots of negative numbers are ordinary. Square roots of negative numbers are not numbers you have — that is the difference to hold on to.
7. **Notes.** A negative exponent is still a reciprocal. The radical does not change a single thing about that.
8. **Example 1 — A power and a radical in one expression.** SPOKEN ONLY.
9. **Example 1 — Worked.** Two independent pieces.
10. **Your Turn — Same steps, your numbers. Boards up when done.** The guide's tiered version: a cube root this time, and a subtraction.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A radical inside a product.** SPOKEN ONLY.
13. **Example 2 — Worked.** Three layers, innermost first: under the bar, then the root, then the multiplication, then the subtraction.
14. **Your Turn — Same steps, your numbers. Boards up when done.** A negative cube root and a product that grows.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. $\sqrt{121}$ → ***11*** *60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]; ±11 — the radical sign names the principal root only [8.NSO.1.7]*

18–19. Board 2. A square patio has an area of 169 square feet. Fencing for it costs 12 dollars a foot. How long is one side of the patio? → ***13 feet*** *The fencing price of 12 dollars a foot is not needed. anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]; 52 — the perimeter, not a side; 84.5 — the area was halved*

20–21. Board 3. $\sqrt[3]{-125}$ → ***−5*** *5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]; 'not a real number' — the square-root rule applied to a cube root [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]*

22–23. Board 4. $\sqrt{2^3 + 17}$ → ***5*** *25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]; anything with two separate roots — the bar groups, so it is one radicand [8.NSO.1.7]*

24–25. Board 5. √(25 + 144). Which is the value? → ***D — 13*** *(A) the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]; (B) the sum was found and the root was never taken [8.NSO.1.7]; (C) the radicand was halved instead of rooted [Unit 2, 2.02: a root is not a half] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. The cube root of −1/64. Which is the value? → ***A — −1/4*** *(B) the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]; (C) the square root of 1/64 was taken and the sign kept [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]; (D) the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $\sqrt{\frac{4}{9}} + \sqrt[3]{\frac{8}{27}}$ → ***4/3*** *12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]; 2/3 — only one of the two radicals was evaluated*

30–31. Board 8. $\sqrt{5^3 - 25} \cdot 10^{-3}$ → ***1/100*** *10,000 — 10^{-3} read as 10^3 [8.NSO.1.3: a negative exponent is a reciprocal]; −1/100 — the exponent's sign made the value's sign [B1G-M 8.NSO.1.3 task: −b vs b^{-1}]*

32–33. Board 9. Simon says that the cube root of −27 is not a real number, because you cannot take the root of a negative. Explain why he is wrong, and give the value. → ***−3*** *'−3' alone — the value without the reason [the item asks why he is wrong]; '3' — the sign dropped as well as the rule misread*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–6 are one root each and walk the cases: a square root, a cube root, a negative radicand, a fraction, a square root with a power under the bar, a cube root with a power under the bar. 7 and 8 each put a power and a radical in one expression — 7 in three shapes of the student's own, 8 in the three the state guide supplies. 9 works backwards in three shapes, including one where the radicand carries a term. The pattern breaks at 10, where the student judges a method rather than a number. 11 is the diagnostic multiple choice on the grouping error and 12 is the select-all, whose two wrong options are the sign of a cube root and an unfinished radicand. The Additional sheet mirrors each position; its question 10 is the same error inside a subtraction.

Misconceptions to Watch

Board 1. 60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]

Board 2. anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]

Board 3. 5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]

Board 4. 25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]

Board 5. the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]

Board 6. the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]

Board 7. 12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]

Board 8. 10,000 — 10^{-3} read as 10^{3} [8.NSO.1.3: a negative exponent is a reciprocal]

Board 9. '−3' alone — the value without the reason [the item asks why he is wrong]

Materials Whiteboards and markers, and Unit 2's reference sheet if students still want the perfect-square list. Calculators are allowed on everything; a calculator will refuse the cube root of a negative on some models, which is worth showing once and explaining rather than trusting.